

Wk2 Alzheimer Report

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R Markdown - Alzheimer Report Output Test

This is an R Markdown document - test italics. Markdown is a simple formatting syntax for authoring **HTML**, **PDF**, and **MS Word** documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

Test:Red text using html code

Test: Red text using LaTeX commands for pdf docs

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

Test Installing pacman packages

```
pacman::p_load(  
  #project and file mgmt  
  here, #file paths relative to R project root folder  
  rio, #import/export of many types of data  
  
  #general data mgmt  
  tidyverse, #several packages for tidy data manipulation  
  
  #plot options  
  viridis, # colour-blind friendly palettes  
  
  #Other  
  dplyr,  
  lubridate,  
  readr,  
  tydr,  
  readxl,  
  janitor,  
  skimr,  
)
```

Test Importing Alz Data

```
#import Alz xlsx data set from BIOS640 Wk 2.1 Exercise
alz_data <- import(here("data", "alzheimers_disease_data.csv"))

#provides dimensions - number of rows and columns
dim(alz_data)
```

```
## [1] 2150 29
```

```
# Shows the structure of data and data types
str(alz_data)
```

```
## 'data.frame': 2150 obs. of 29 variables:
## $ PatientID : int 4751 4752 4753 4754 4755 4756 4757 4758 4759 4760 ...
## $ Age : chr "\"73\"\"" "\"89\"\"" "\"73\"\"" "\"74\"\"" ...
## $ Gender : int 0 0 0 1 0 1 0 0 1 0 ...
## $ Ethnicity : int 0 0 3 0 0 1 3 0 1 0 ...
## $ Education : int 2 0 1 1 0 1 2 1 0 0 ...
## $ BMI : num 22.9 26.8 17.8 33.8 20.7 ...
## $ Smoking : int 0 0 0 1 0 0 1 0 0 1 ...
## $ Alcohol : num 13.3 4.54 19.56 12.21 18.45 ...
## $ PhysicalActivity: num 6.33 7.62 7.84 8.43 6.31 ...
## $ DietQuality : num 1.347 0.519 1.826 7.436 0.795 ...
## $ SleepQuality : num 9.03 7.15 9.67 8.39 5.6 ...
## $ FamilyHistoryAD : int 0 0 1 0 0 0 0 0 0 0 ...
## $ CVD : int 0 0 0 0 0 0 0 0 0 1 ...
## $ Diabetes : int 1 0 0 0 0 1 0 0 0 0 ...
## $ Depression : int 1 0 0 0 0 0 0 0 0 0 ...
## $ HeadInjury : int 0 0 0 0 0 0 1 0 0 0 ...
## $ Hypertension : int 0 0 0 0 0 0 0 0 1 0 ...
## $ mmse : num 17.8 10.34 23.61 22.21 9.21 ...
## $ Functional : num 7.03 3.8 6.65 4.68 3.68 ...
## $ Memory : int 0 0 1 0 1 1 0 1 0 0 ...
## $ Behavioral : int 0 0 1 0 1 0 0 0 0 1 ...
## $ ADL : num 6.77 3.45 5.41 6.13 2.48 ...
## $ Confusion : int 0 0 0 0 0 1 0 1 0 0 ...
## $ Disorientation : int 0 0 0 0 0 1 0 0 0 0 ...
## $ Personality : int 0 0 0 0 1 0 0 0 1 0 ...
## $ Tasks : int 1 0 1 0 1 0 0 0 0 0 ...
## $ Forgetfulness : int 0 0 0 0 0 1 0 0 1 0 ...
## $ Diagnosis : int 0 0 0 0 0 0 0 1 0 0 ...
## $ Hospital : chr "Hospital C" "Hospital D" "Hospital F" "Hospital D" ...
```

Test Cleaning names

```
# automatically clean column names using the clean_names() function from the janitor package
# load package
pacman::p_load(janitor)

# clean names - i.e., lowercase and underscores
alzheimer_data <- alz_data %>%
```

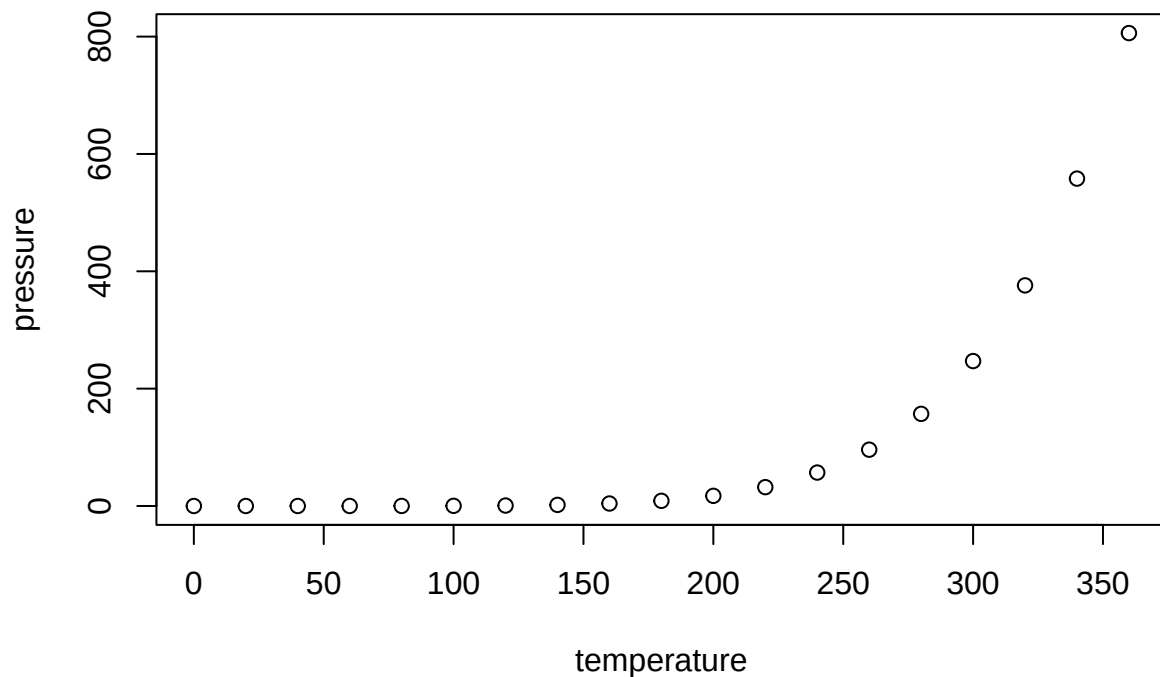
```
clean_names()

names(alzheimer_data)

## [1] "patient_id"      "age"              "gender"
## [4] "ethnicity"       "education"        "bmi"
## [7] "smoking"         "alcohol"          "physical_activity"
## [10] "diet_quality"    "sleep_quality"    "family_history_ad"
## [13] "cvd"             "diabetes"         "depression"
## [16] "head_injury"     "hypertension"     "mmse"
## [19] "functional"      "memory"           "behavioral"
## [22] "adl"             "confusion"        "disorientation"
## [25] "personality"     "tasks"            "forgetfulness"
## [28] "diagnosis"       "hospital"
```

Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.

Test third level Heading - Bullet Points

Here are bullet points where you need to leave an empty space after the colon and before the first bullet:

- Bullet1
- Bullet2
- sub-bullet1
- Sub-bullet2
- Bullet3

Test Comment Out

The text below this should not be visible in the output.

Testing inline math expression The mean is $\bar{x}$

Test Code Chunks below

```
# your R code here
mean(alzheimer_data$age)
```

```
## [1] NA
```

Test Plot “{rage-plot, echo = FALSE} plot(alzheimer_data\$age)

““

Test Markdown Table ## Ethnicity Summary

Statistic	African American	Asian	European descent	Other
Count	454	206	1278	211

Test Image insertion

